A3_CP414_W2026

Sunday, March 8, 2026 2:02 PM

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CP-414 Winter 2026 Assignment #3
Due: Tuesday, March 16

1. Consider the context free grammar **G** with the following rules (**S** is start variable, **A** and **B** are variables, **a** and **b** are terminals):

   S → A | B | b
   A → aSbS
   B → aS

   Show that this grammar is ambiguous by giving an example of a string in the language that has **two different parse trees**. Show your work (in particular, **show both parse trees**). Try to find as short ambiguous string as possible.

2. Convert **G** from question 1 to an equivalent PDA using the procedure given in your text and class.

3. Convert the following CFG (**A is start variable, 0 and 1 - terminals**) into an equivalent CFG in Chomsky normal form, using the procedure given in the class. **Show all intermediate steps**.

   A → ABC | AC | B
   C → BC1 | B
   B → 00 | ε

4. Give a context-free grammar generating the language of all strings over alphabet **{0,1}** with the number of **0**s equal to the number of **1**s plus 2. For example, strings 010100, 110000, 0100, 1000, 010001 are in this language, while strings 0110, 1, 0, 0000000001 are not. **Justify correctness of your grammar.**

5. Consider the two languages: L1 = $\{x^n y^m x^m z^n \mid n > 0, m > 0\}$ and L2 = $\{x^n y^n x^n z^m \mid n > 0, m > 0\}$ over alphabet {x, y, z}. Show that one of them is context-free (by providing a context-free grammar that defines this language) and another is not context-free (by using the Pumping Lemma for context free languages).

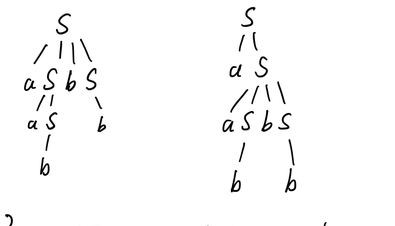
6. Let L be the language of all palindromes over **{a, b}** containing equal numbers of **a**'s and **b**'s. Show that L is not context free using Pumping Lemma for context-free languages.

7. (**bonus**) Construct the CFG and PDA for the languages given by:

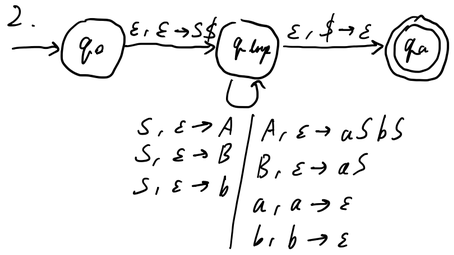
   a. $L = \{wa^n w^R \mid w \in \{a, b\}^*, n \geq 0\}$
   b. $L = \{(01)^k 0^k \mid k \geq 0, \Sigma = \{0,1\}\}$

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1. $s = aabbb$ is ambiguous



2.



3. $S \to A$ — Add start S
   $A \to ABC \mid AC \mid B$
   $C \to BC1 \mid B$
   $B \to 00 \mid \varepsilon$

   $S \to A \mid \varepsilon$ — Remove ε-rule
   $A \to ABC \mid AC \mid B \mid \varepsilon \mid AB \mid A \mid BC \mid C$
   $C \to BC1 \mid B \mid C1 \mid \varepsilon \mid B1 \mid 1$
   $B \to 00 \mid \varepsilon$

   $S \to A \mid \varepsilon$ — Remove unit-rule
   $A \to ABC \mid AC \mid B \mid AB \mid A \mid BC \mid C \mid BC1 \mid C1 \mid B1 \mid 1 \mid 00$
   $C \to BC1 \mid B \mid C1 \mid B1 \mid 1 \mid 00$
   $B \to 00$

   $S \to A \mid \varepsilon$ — Add binary-rule
   $A \to ABC \mid AC \mid AB \mid BC \mid BC1 \mid C1 \mid B1 \mid 1 \mid 00 \mid BC_0 \mid AB_0$
   $C \to BC1 \mid C1 \mid B1 \mid 1 \mid 00 \mid BC_0$
   $B \to 00$
   $C_0 \to C1$
   $B_0 \to BC$

   $S \to A \mid \varepsilon$ — Remove mixed-rule
   $A \to AC \mid AB \mid BC \mid C1 \mid B1 \mid 1 \mid 00 \mid BC_0 \mid AB_0 \mid CU_1 \mid BU_1 \mid U_0U_0$
   $C \to C1 \mid B1 \mid 1 \mid 00 \mid BC_0 \mid CU_1 \mid BU_1 \mid U_0U_0$
   $C_0 \to C1 \mid CU_1$
   $B_0 \to BC$
   $B \to 00 \mid U_0U_0$
   $U_1 \to 1$
   $U_0 \to 0$

   CNF:
   $S \to A \mid \varepsilon$
   $A \to AC \mid AB \mid BC \mid BC_0 \mid AB_0 \mid CU_1 \mid BU_1 \mid U_0U_0 \mid 1$
   $C \to BC_0 \mid CU_1 \mid BU_1 \mid U_0U_0 \mid 1$
   $C_0 \to CU_1$
   $B_0 \to BC$
   $B \to U_0U_0$
   $U_1 \to 1$
   $U_0 \to 0$

4. $S \to TOTOT$
   $T \to OTIT \mid ITOT$

   The first rule guarantees that:
   - string cannot be ε
   - There are 2 0s in string

   The second rule guarantees that:
   - $T = \{(0|1)^* \mid \#0 = \#1\}$

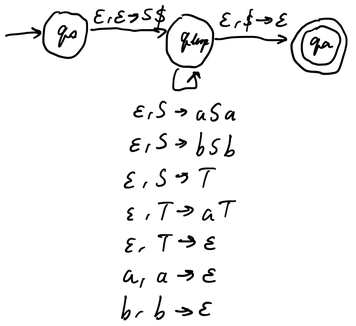
   Together:
   - There are 2 more 0s than 1s
   - Anywhere surrounding the 2 extra 0s, any number of 0 and 1s where $\#0 = \#1$ are possible.

5. CFG of L1:
   $S \to xSz \mid xAz$
   $A \to yAx \mid yBx$
   $B \to \varepsilon$

   L2 Proof:
   Let p be pumping length, choose $s = x^p y^p x^p z^p$, such that $s \in L2$, $|s| > p$.
   Since pumping up only increases xy or yx ($|vxy| \leq p$), but not xyz in the sametime, s cannot be pumped.
   By contradiction, L1 ∉ CFL.

6. Let p be pumping length, choose $s = b^{p/2} a^p b^{p/2}$, such that $s \in L$, $|s| > p$. Pump up.
   Case 1:
   $b^{p/2} a^{p+k} b^{p/2}$, $b^{p/2+k} a^p b^{p/2}$, $b^{p/2} a^p b^{p/2+k} \notin L$, $k > 0$
   since $\#a \neq \#b$.
   Case 2:
   $b^{p/2+k} a^{p+k} b^{p/2}$, $b^{p/2} a^{p+k} b^{p/2+k} \notin L$, $k > 0$
   since it breaks symmetry (i.e. not palindrome).
   s cannot be pumped.
   By contradiction, L ∉ CFL.

7. a. $S \to aSa \mid bSb \mid T$
      $T \to aT \mid \varepsilon$



   b. $S \to 0IS0 \mid \varepsilon$

